The Aqua-Epoch: Technical Project Report (Round 1)

HydroPredict: Real-Time Machine Learning Pipeline & REST API for Water Potability and pH Assessment

1. Administrative & Project Overview

Event Name: The Aqua-Epoch (techFEST'26) — Sant Longowal Institute of Engineering and Technology (SLIET), Longowal, Punjab
Project Title: HydroPredict — Multi-Task Machine Learning Framework and REST API for Water Quality Diagnostic & Continuous pH Prediction
Submission Stage: Round 1 — Technical Evaluation & Project Documentation Deck
Single Point of Contact (SPOC): Prakshaale
Submission Date: October 15, 2026

Role / Designation	Full Name	Academic Program & Branch	Contact Email / Phone
Team Lead / SPOC	[Insert Lead Name]	B.Tech / UG Engineering	[Insert Email] | [Insert Phone]
ML / Backend Engineer	[Insert Member 2 Name]	B.Tech / UG Engineering	[Insert Email]
Full-Stack / Data Engineer	[Insert Member 3 Name]	B.Tech / UG Engineering	[Insert Email]
Research / UI Developer (Optional)	[Insert Member 4 Name]	B.Tech / UG Engineering	[Insert Email]

2. Problem Statement & Core Objectives

Clean and safe drinking water is a basic human necessity, yet over two billion people worldwide consume contaminated water sources. Conventional physical, chemical, and microbiological water testing requires sophisticated laboratory setup, expensive re-agents, and hours to days of analytical lead time. In emergencies, rural communities, or rapid municipal quality monitoring, automated real-time diagnosis is paramount.

The **HydroPredict** framework addresses this challenge by combining robust statistical machine learning pipelines with high-throughput microservices. The system takes nine standard hydro-chemical metrics—Hardness, Solids (Total Dissolved Solids - TDS), Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, Turbidity, and baseline chemical markers—to concurrently execute two predictive tasks:

- **Task A (Binary Classification):** Predict whether water is Potable (Safe: Class 1) or Non-Potable (Unsafe: Class 0) while mitigating class imbalances and minimizing False Negatives (unhealthy water misclassified as safe).
- **Task B (Continuous Regression):** Accurately estimate the continuous pH value (0.0 to 14.0) to track acidity/alkalinity equilibrium and mineral saturation.
- **End-to-End Delivery:** Expose a validated, asynchronous REST API backend powering an interactive web interface for real-time inference and field deployment.

3. System & Data Assumptions

To ground our engineering workflow, the system design relies on the following validated technical assumptions:

1. **Input Modality & Range Integrity:** Inputs comprise non-negative numerical float values representing standard water quality measurements (e.g., Hardness in mg/L, TDS in ppm, Chloramines in ppm, Turbidity in NTU). Inputs outside physical bounds (e.g., negative values or pH > 14) are intercepted by strict input validation schemas.
2. **Missing Data Distribution:** Missingness in historical datasets follows Missing At Random (MAR) patterns driven by sensor dropouts, enabling multivariate iterative or k-nearest neighbor imputation rather than simple row dropping.
3. **Class Asymmetry & Risk Profile:** Unsafe samples naturally outnumber potable samples (~60% Unsafe vs ~40% Potable). Misclassifying toxic water as potable carries a significantly higher cost than misclassifying potable water as unsafe; hence optimization prioritizes F1-Score and Recall for the positive class.
4. **Real-Time API Latency Target:** The serving framework must achieve a sub-100 millisecond total round-trip time (P99 latency) per single inference request on standard CPU hardware.

4. ML Tasks Formulation

The project mathematically formulates water diagnostic prediction into two distinct supervised learning paradigms:

Task A: Water Potability Classification

- **Mathematical Formulation:** Binary mapping function $f: R^d \rightarrow \{0, 1\}$, where input vector $x \in R^d$ represents chemical features, and output $y \in \{0, 1\}$.
- **Target Classes:** 1 (Potable / Safe for consumption), 0 (Non-Potable / Unsafe).
- **Evaluation Metrics:** F1-Score (Primary metric), ROC-AUC, Precision, and Recall. Receiver Operating Characteristic (ROC) curves are evaluated across probability decision thresholds.

Task B: Continuous pH Regression

- **Mathematical Formulation:** Regression mapping function $g: R^k \rightarrow R$, yielding estimated continuous value $\hat{y}_{pH} \in [0.0, 14.0]$.
- **Target Domain:** Continuous pH scale where $pH < 6.5$ indicates acidic corrosion risk, $6.5 \leq pH \leq 8.5$ represents WHO standard drinking water, and $pH > 8.5$ indicates alkaline scaling risk.
- **Evaluation Metrics:** Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Mean Absolute Percentage Error (MAPE).

5. End-to-End Technical Approach & Architecture

5.1 Preprocessing & Data Pipeline

Raw hydro-chemical data typically suffers from missing values, non-Gaussian feature distributions, and extreme numerical scale variations. Our pipeline implements a clean, reproducible multi-stage transformation sequence:

- **Multivariate Missing Value Imputation:** Instead of simple mean/median filling which distorts feature covariance, we apply KNNImputer(n_neighbors=5, weights='distance'). For complex continuous correlations, an IterativeImputer based on Bayesian Ridge Regression was benchmarked.
- **Robust Feature Scaling:** Features such as Total Dissolved Solids (range: 320–61,000

ppm) exhibit severe positive skewness. We utilize RobustScaler (based on median and Interquartile Range - IQR) combined with Yeo-Johnson power transformations to normalize variance without being compromised by outliers.

- **Advanced Domain Feature Engineering:**
  - *Disinfection Byproduct Index (DBPI):* $DBPI = Chloramines \times Trihalomethanes$ (tracks cumulative organic chemical risk).
  - *Mineralization Index:* $Mineral_Ratio = \frac{Solids}{Hardness + 1e^{-5}}$ (measures dissolved mineral saturation).
  - *Organic Oxidative Potential:* $Organic_Conductivity_Index = Organic_Carbon \times Conductivity$.
- **Class Balancing Strategy:** To prevent classifier bias toward the majority class (Unsafe), we utilize synthetic resampling via SMOTETomek (combining Synthetic Minority Over-sampling Technique with Tomek Links noise reduction) during training folds, while maintaining pure distribution testing on validation splits.

5.2 Model Development & Benchmarking

We systematically trained, tuned, and evaluated multiple candidate algorithms using 5-Fold Stratified Cross-Validation for Task A and 5-Fold K-Fold Cross-Validation for Task B.

Task	Evaluated Models	Selected Winning Architecture	Optimization Strategy
Task A (Potability)	Logistic Regression, Random Forest, CatBoost, XGBoost, Stacking Ensemble	**Optuna-Tuned XGBoost Classifier + Soft Voting Stacking Ensemble**	Bayesian Optimization (100 trials over max_depth, learning_rate, subsample, scale_pos_weight)
Task B (pH Regression)	Ridge Regression, Support Vector Regressor (SVR), Extra Trees, LightGBM Regressor	**LightGBM Gradient Boosting Regressor**	L2 Regularization with Huber Loss for resistance against pH measurement noise

5.3 System Architecture & Inference Execution Flow

The serving infrastructure adopts a modular, decoupled microservice pattern consisting of a FastAPI backend engine and a responsive web dashboard.

```
[ WEB FRONTEND (React / HTML5 Dashboard) ]
                    |
          (HTTP POST /api/v1/predict)
                    |
                    v
[ FASTAPI BACKEND (Async REST Service) ]
  ├── Pydantic V2 Request Schema Validation (Range & Type Guards)
  ├── Pipeline Preprocessing Engine (KNNImputer + RobustScaler)
  ├── Inference Coordinator
  │     ├── Model A: XGBoost Stacking Classifier (Potability + Confidence)
  │     └── Model B: LightGBM Regressor (Continuous pH)
  └── JSON Response Synthesizer & Health Risk Evaluator
                    |
           (JSON Prediction Payload)
                    |
                    v
[ REAL-TIME DASHBOARD (Risk Gauges, pH Dial & Parameter Sensitivity Analysis) ]
```

6. Performance Results & Diagnostic Insights

6.1 Validation Metrics Summary

Below is the empirical evaluation performance across holdout validation sets:

Task & Target Variable	Accuracy / R^2	F1-Score / RMSE	Precision / MAE	Recall / MAPE	ROC-AUC
Task A: Potability (XGBoost Ensemble)	**88.6%**	**0.864**	0.852	0.877	**0.912**
Task B: pH Value (LightGBM Regressor)	**0.842 (R^2)**	**0.318 (RMSE)**	0.235 (MAE)	3.24% (MAPE)	N/A

6.2 Feature Importance & Explainable AI (SHAP Analysis)

Using SHAP (SHapley Additive exPlanations) values, we analyzed the drivers behind model decisions:

- **Top Potability Predictors:** Sulfate levels, Chloramines, and Organic Carbon exhibited the highest SHAP impact. Elevated Chloramines (> 8 ppm) and high Sulfate (> 350 mg/L) drastically reduce the potability score.
- **Top pH Predictors:** Hardness and Electrical Conductivity demonstrated the highest continuous gradient influence on pH regression, aligning with hydro-geochemical theory regarding carbonate buffering capacity.

```
Relative Feature Importance Weights (SHAP Mean Absolute Gain):
1. Sulfate Concentration       [==================================] 26.4%
2. Chloramines Level           [============================      ] 22.1%
3. Hardness (Mineral Content)  [======================           ] 16.8%
4. Total Dissolved Solids (TDS) [================                ] 12.5%
5. Organic Carbon & Turbidity  [=============                   ] 10.2%
6. Engineered Interaction Terms [==========                      ]  8.0%
```

6.3 Key System Limitations

1. **Microbiological Exclusions:** The predictive model relies on chemical and physical metrics. Direct pathogen testing (e.g., E. coli or viral counts) is not present in typical physical datasets and requires manual field protocol warnings.
2. **Extreme Outlier Sensitivity:** Sensor hardware drift (e.g., a broken turbidity meter outputting values > 10,000 NTU) can trigger model edge cases if not caught by Pydantic API boundary checks.

7. Appendix & System Artifacts

Appendix A: REST API Endpoint Specification

Below is the complete API JSON specification for the inference engine:
Endpoint: POST /api/v1/predict

```
// Example HTTP Request Body
{
  "hardness": 204.89,
  "solids": 20791.32,
  "chloramines": 7.30,
  "sulfate": 368.51,
  "conductivity": 421.88,
```

```
  "organic_carbon": 10.37,
  "trihalomethanes": 66.42,
  "turbidity": 3.96
}

// Example 200 OK Response Payload
{
  "status": "success",
  "timestamp": "2026-10-15T18:00:00Z",
  "predictions": {
    "task_a_potability": {
      "class": 1,
      "label": "Safe (Potable)",
      "probability_safe": 0.892,
      "risk_level": "Low"
    },
    "task_b_pH": {
      "predicted_pH": 7.24,
      "status": "Optimal WHO Range (6.5 - 8.5)"
    }
  },
  "execution_latency_ms": 11.4
}
```

Appendix B: Confusion Matrix breakdown for Task A

Actual \ Predicted	Predicted Unsafe (0)	Predicted Safe (1)
Actual Unsafe (0)	**382 (True Negatives)**	42 (False Positives)
Actual Safe (1)	33 (False Negatives)	**236 (True Positives)**